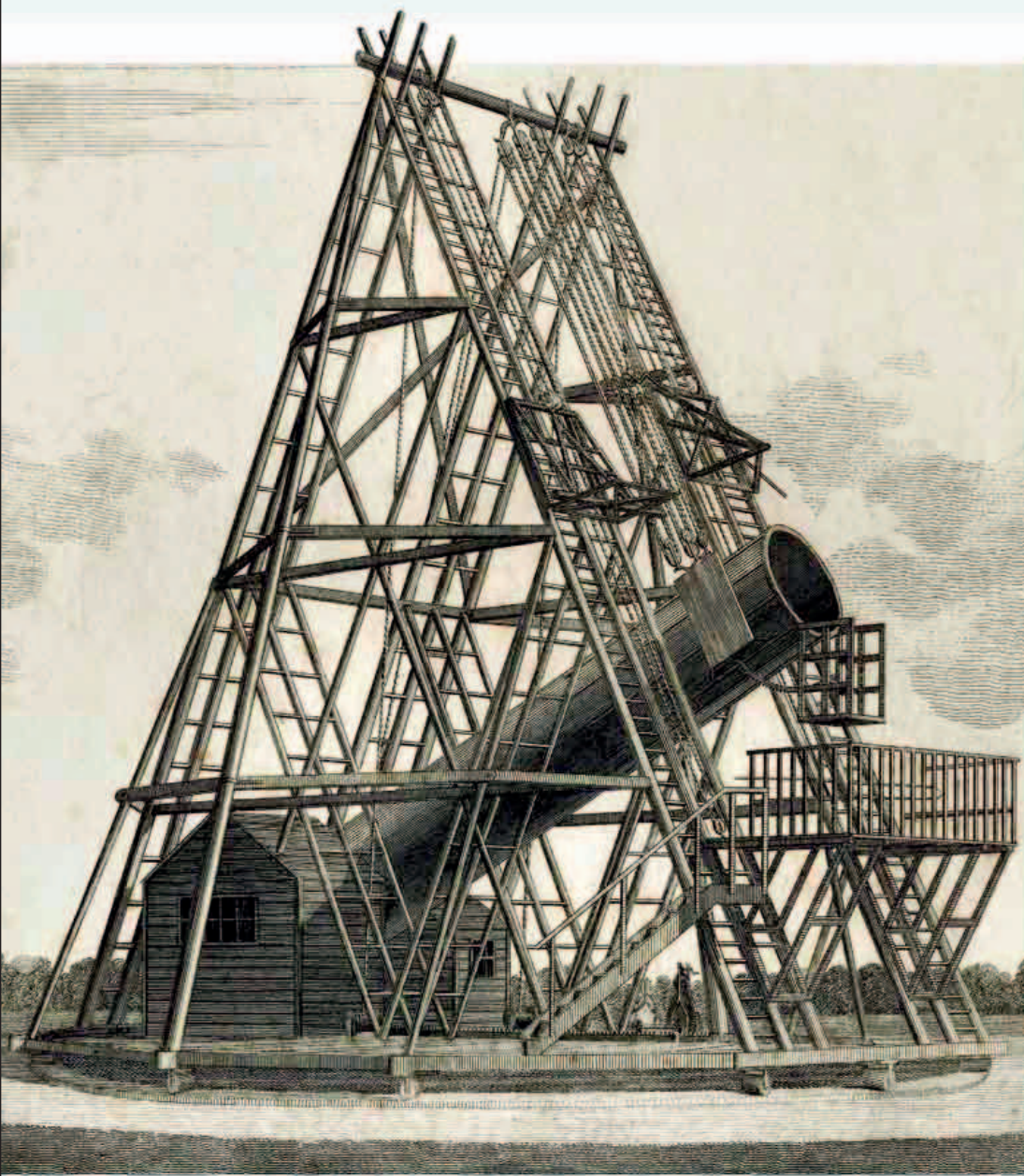


“...WE MUST ADMIT, **AS ELEMENTS**, ALL THE **SUBSTANCES INTO WHICH WE ARE CAPABLE**, BY ANY MEANS, TO REDUCE BODIES **BY DECOMPOSITION**.”

Antoine Lavoisier, French chemist, from *Traité élémentaire de chimie* (*Elements of Chemistry*), 1789



AGAINST THE BACKDROP OF THE FRENCH REVOLUTION,

a scientific revolution was underway in 1789. French chemist **Antoine Lavoisier** completed his *Elements of Chemistry*, a book that laid the foundations of **chemistry as a science**. Lavoisier created the first table of elements, which also included heat and light. He introduced international chemical symbols and introduced modern names, such as oxygen and hydrogen for gases and sulphates for compounds.

On 28 August, German-born British astronomer **William Herschel** looked through his **largest telescope** for the first time. He had already built several telescopes, but this 40ft (12m) reflecting telescope was the biggest. That night, Herschel discovered a **new moon of Saturn, Enceladus**, and three weeks later he spotted another, **Mimas**.

French botanist **Antoine Laurent de Jussieu** published a system for classifying flowering plants. It used Linnaeus's two-part Latin naming system (see 1754) and divided flowering plants

Herschel's telescope

Astronomer William Herschel took four years to build this giant telescope in his garden in Slough, England. It was paid for by King George III, and dismantled in 1839.



ANTOINE LAVOISIER
[1743–94]

Born in Paris, France, Antoine Lavoisier is sometimes said to be the father of chemistry—he proved the role of oxygen in combustion, which laid the foundation for modern chemistry. Lavoisier established that matter is neither created nor destroyed during chemical changes. A target for revolutionaries, he was guillotined on May 8, 1794.

into classes according to the number of stamens and pistils. This system is still used today.

In Canada, British explorer **Alexander Mackenzie** (1764–1820) set out in his canoe, following the uncharted 1,100 mile (1,770 km) river—now known as Mackenzie River—to the Arctic Ocean.

French chemist **Antoine Lavoisier** completes *Elements of Chemistry*

British naturalist **Erasmus Darwin** publishes his poem, *The Loves of the Plants*

August 28 British astronomer **William Herschel** discovers **Enceladus**, a new moon of Saturn

French botanist **Antoine Laurent de Jussieu** publishes a system for classifying flowering plants

July 14 British explorer **Alexander Mackenzie** reaches the Arctic Ocean

September 17 **William Herschel** discovers Saturn's moon **Mimas**